

Pass@1 cannot distinguish reasoning quality from answer correctness

Two real examples where accuracy metrics give misleading signals (full model outputs in Appendix B)

(a) Correct Answer, Poor Reasoning

CORRECT

Gemma-7B | MATH500 #452 | GT: 143

Q: The sum of the digits of a two-digit number is 13. The difference between the number and the number with its digits reversed is 27. What is the sum of the original number and the number with its digits reversed?

Key output excerpts:

$x+y=13$ and $(10x+y)-(10y+x)=27$

$9x=14$, $x=14/9=1.5$

"x must be integer, so $x=1$ " -> $y=12$

invalid digit

Number= $10(1)+12=22$; Reversed= $10(12)+1=121$

Sum = $22+121 = 143$ -> $\boxed{143}$

correct by coincidence

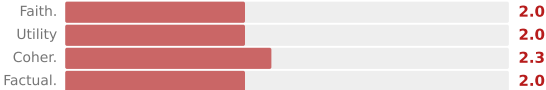
Pass@1

1

Reasoning

2.1 / 5

Misleading success



(b) Wrong Answer, Strong Reasoning

INCORRECT

Phi-4-Reasoning | GSM8K #242 | GT: 3

Q: Mike was a pen pal with 5 people. He stopped being penpals with 2 of them. They each send 2 letters a week that are 5 pages long. He responds in kind. He can write a page every 6 minutes. How many hours does he spend writing a week?

Key output excerpts:

3 pals x 2 letters = 6 letters, $6 \times 5 = 30$ pages
He writes 30 pages. $30 \times 6\text{min} = 180\text{min} = 3$ hrs

<think> "Final answer: $\boxed{3}$ hours."

correct CoT

</think> "Combined, he writes $30+30 = 60$ pages"

$60 \times 6\text{min} = 360\text{min} = 6$ hrs -> $\boxed{6}$

double-count

Pass@1

0

Reasoning

4.1 / 5

Hidden capability

